Cross-Cube Vega Hedging

A Swaption Market-Making Lab

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Independent research - produced as a portfolio artefact

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Abstract and Executive Summary

Abstract

A swaption market-maker sees vega by cube point, but the realised cube does not move one point at a time. It moves through correlated factors: ATM level, expiry slope, curvature, and skew-wing dynamics. This lab simulates that cube, calibrates noisy SABR smiles, runs a client-flow market-maker, and compares four hedging policies. The factor-neutral rule compresses terminal P&L dispersion and improves the left tail. The limitation is explicit: no live market data, fixed beta, daily hedging, and a quoting overlay that is a factor-space heuristic rather than a full HJB solution.

Executive Summary

Single-tenor vega is a mirage. It is a convenient control number, but it is not the basis in which the cube actually moves. The same-axis terminal P&L overlay below is the central exhibit: delta-only hedging leaves a fat distribution, while factor-neutral hedging concentrates the outcome around a tighter risk budget. The point is not that factor-neutral hedging is free. It trades more, pays more hedge cost, and depends on liquid proxy instruments. The point is that those costs buy a cleaner residual.

Strategy	Mean	Std	5% VaR	Sharpe
Delta-only	30.7	70.6	-77.5	0.43
Factor-neutral	9.7	6.4	-0.5	1.52

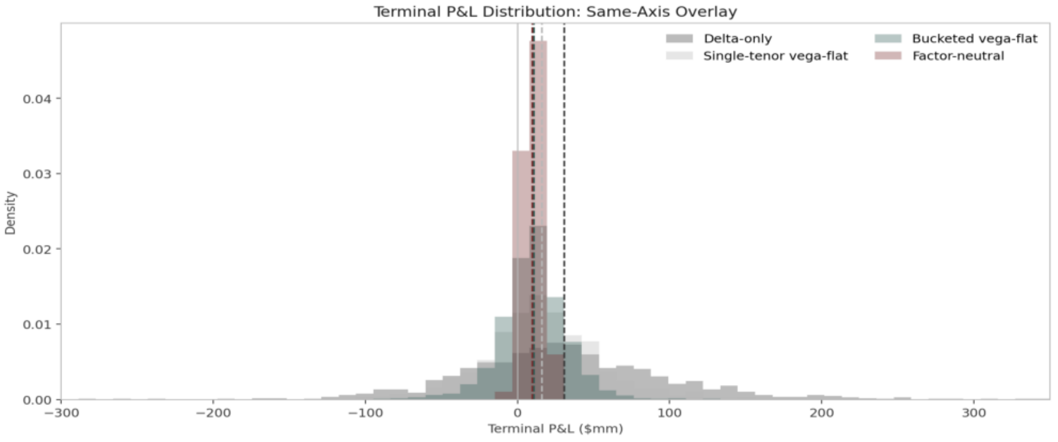


Figure 1. Same-axis terminal P&L overlay; dashed lines mark strategy means.

Introduction and Motivation

Why Node Vega Exists

Node vega is the reporting standard because it is operationally legible. Traders quote points, risk systems aggregate points, and end-of-day controls ask whether a trader is long or short a recognisable part of the cube. For a desk, that convention is useful. For a hedge, it is incomplete. The cube does not wait for one node to move; the whole surface breathes through common shocks.

The Risk Basis

A block in 2y into 10y exposes the book to more than the 2y x 10y mark. It has projection onto level, onto the front-back expiry slope, onto an intermediate-expiry hump, and onto skew and wing dynamics. A local vega hedge can be flat at the booked point and still be long the level factor. That is exactly the leakage this project tries to isolate.

The Bartlett Pattern

Bartlett's SABR delta correction is a useful mental model. The Black delta is the obvious hedge only if volatility parameters stand still. Once alpha co-moves with the forward, the hedge needs the parameter response too. Cross-cube vega has the same shape: the obvious node hedge ignores the way nearby smiles move together.

Trader's-Day Framing

The market-maker takes down flow, earns edge, and goes home with an inventory. The next morning the cube has moved. The question is not whether the single node was hedged, but whether the hedge was short the same factor the book was long. If it was not, the desk has paid spread for a residual it did not intend to warehouse.

Architecture

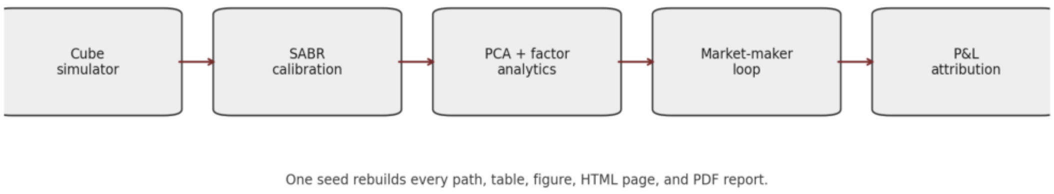


Figure 2. Deterministic pipeline: simulator, calibration, analytics, market-maker, P&L.

Pipeline Walk-Through

The simulator produces a full SABR cube path: forwards, annuities, strikes, ATM vols, rho, nu, alpha, and quoted smile vols. Calibration consumes a noisy snapshot of those market vols and writes residual heatmaps plus static-arbitrage counts. PCA analytics recover the latent risk basis. The market-maker loop samples client flow, applies the quoting overlay, hedges at end-of-day, and emits path-level P&L. Attribution then reconciles edge, theta, delta, first-order vega, cross-vega, and hedge cost.

Reproducibility Contract

The master seed sits in YAML. The observation-noise stream is separate from the simulator streams, so calibrating a noisy market snapshot does not perturb the true cube path. The Makefile rebuilds simulations, figures, the static site, and this PDF from scratch.

Model: True Cube I

ATM Factor Surface

The cube is indexed by option expiry T in {1m, 3m, 6m, 1y, 2y, 5y, 10y} and underlying swap tenor tau in {1y, 2y, 5y, 10y}. The ATM Black-vol surface is driven by three smooth loadings over this grid.

$$\log \sigma_{ATM}(T, \tau, t) = \mu(T, \tau) + \sum_i L_i(T, \tau) X_i(t)$$

Loading Shapes

The level loading is nearly constant across the cube with a small tenor tilt. The slope loading is the centred log-expiry coordinate. The curvature loading is a Gaussian hump around the intermediate expiries, centred near two years, then normalised. These are not calibrated to live data; they are deliberately simple shapes that a trader would recognise from daily vol-surface moves.

OU Dynamics

$$dX_i = -\kappa_i X_i \, dt + \eta_i \, dW_i, \quad corr(dW) = R$$

The OU factors mean-revert daily. Their instantaneous covariance is configurable in YAML, and the market-maker quoting overlay uses the full covariance matrix, including off-diagonal terms. That matters because level inventory and skew inventory should interact rather than widen quotes independently.

Stylised Scale

The front of the normal-vol cube is set around the 60-90bp region and the back is anchored near 80bp, consistent with the broad scale used in interest-rate volatility modelling examples. Absolute levels are laboratory choices, not market claims.

Model: True Cube II

Skew and Wing Dynamics

Each expiry-tenor slice carries SABR parameters alpha, beta, rho, and nu. Beta is fixed at the configured value. Rho is generated through a transformed OU process, so it stays inside [-0.95, 0.0]. Log-nu is another OU process. Both shocks are correlated with the level factor, reflecting the stylised observation that skew and volatility level do not move independently.

ATM Inversion

The simulator specifies ATM vol first, then recovers alpha from Hagan's ATM SABR formula. This keeps the factor model attached to the observable surface rather than to an abstract alpha process. Calibration mirrors market practice by enforcing the ATM point and fitting smile shape through rho and nu.

SABR and Vol Conversion

The pricer implements Hagan's 2002 lognormal SABR formula with a shifted extension for negative-rate laboratory cases. Black-76 and Bachelier pricing are both present. Normal/lognormal conversion is done by Brent root-finding on price equality rather than by a closed-form approximation; the extra cost is small and the audit trail is clean.

Parameter Honesty

The configuration file is not a calibration file. It records a plausible set of stylised facts: mean-reverting factors, correlated level/skew shocks, bounded rho, and liquid hedge points. The report should be read as a controlled experiment, not as a claim about today's swaption market.

Model: Cube Snapshots

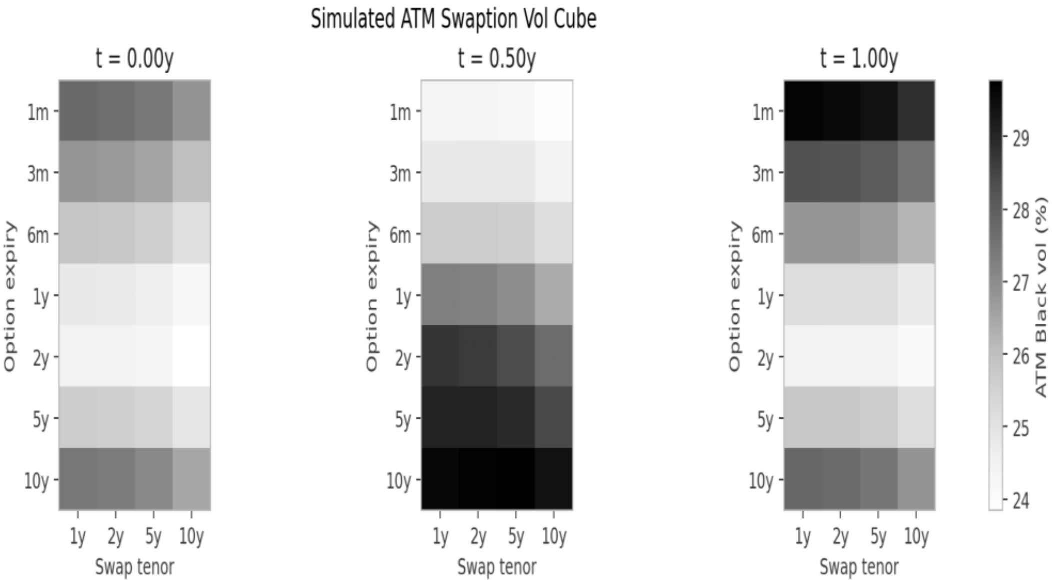


Figure 3. Three ATM cube snapshots from the simulated path.

Reading the Heatmaps

The snapshots are deliberately boring in the right way: the whole cube changes, and nearby expiries and tenors change together. A single isolated node shock would make the hedging problem artificial. The thesis needs a cube that moves in common modes, because that is where local vega reports become a lossy compression.

Calibration I

Noisy Market Snapshot

The calibration target is no longer the exact data-generating SABR surface. Each strike vol receives independent multiplicative lognormal observation noise: sigma_market = sigma_true exp(eps), eps ~ N(0, sigma_obs^2), with sigma_obs = 0.005 by default. The random stream is separate from the simulator stream.

Per-Slice Fit

Per-slice calibration fixes beta, matches ATM exactly through alpha inversion, and fits rho and nu by least squares. That is close to the workflow a desk would use when bootstrapping individual smile slices. It gives low local residuals, but it has no reason to respect calendar consistency across neighbouring expiries.

Cube-Constrained Fit

The joint objective keeps the same ATM discipline and adds penalties for butterfly and calendar violations, plus a smaller smoothness penalty on rho and log-nu over the cube. The no-arbitrage penalties dominate smoothness by design. This is not a local least-squares beauty contest; it is a risk-control fit.

$$J = ||\sigma_m - \sigma_q||^2 + \lambda_b B + \lambda_c C + \lambda_s S$$

The diagnostic question is whether a small residual cost buys fewer static-arbitrage breaks. If it does not, the constrained calibration story fails. The table on the next page is therefore a core acceptance test, not decoration.

Calibration II

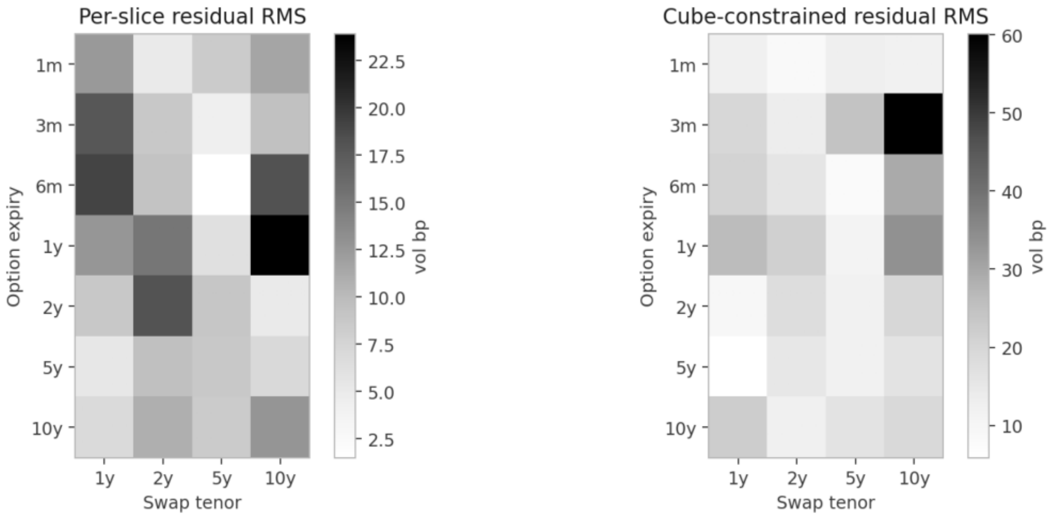


Figure 4. Residual RMS in vol bp. Joint calibration pays local residual for consistency.

Static-Arbitrage Counts

Fit	Total	Butterfly
Per-slice	3	3
Cube-constrained	0	0

The noisy per-slice fit now has genuine residuals instead of a meaningless 1e-12 colourbar. The constrained fit is slightly worse locally, but it removes the material calendar breaks under the same checker. This is the calibration analogue of the hedging thesis: a local optimum can be the wrong portfolio object.

Factor Recovery via PCA I

Method

PCA is run on daily changes in the simulated ATM log-vol surface. The first three components explain the designed level, slope, and curvature subspace. Because PCA basis vectors are sign-ambiguous, the diagnostic aligns them to the known simulator loadings and flips signs consistently.

Cosine Similarity

Factor	Cosine	Expl. var
Level	1.000	59.8%
Slope	1.000	32.8%
Curvature	1.000	7.3%

What This Validates

The table is not claiming that PCA can name economic factors in real data without judgement. It says something narrower and important: in this controlled cube, the analytics recover the latent risk subspace used by the hedger. That is enough to make the hedging comparison internally coherent.

What It Does Not Validate

It does not validate the parameter levels, the absence of jumps, or the stability of factor loadings through regimes. A live implementation would rerun this diagnostic over rolling windows and ask whether factor labels are stable enough to trade.

Factor Recovery via PCA II

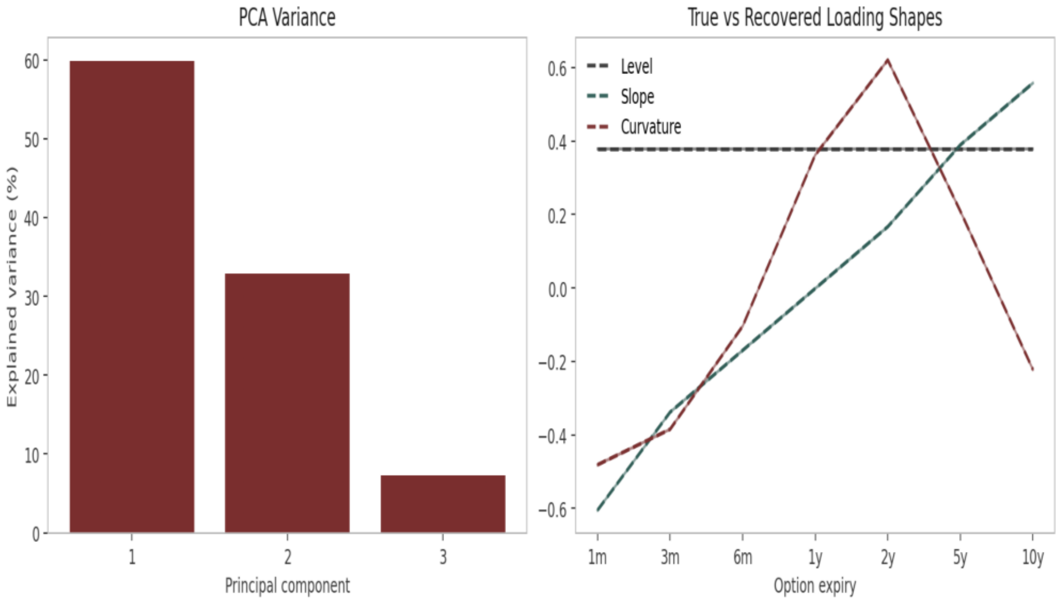


Figure 5. Dashed true loadings and solid recovered loadings are normalised consistently.

Reading the Figure

The recovered curves are plotted against the same expiry axis and colour as the true factor labels. The diagnostic was previously misleading because the true loadings and recovered eigenvectors lived on different scales. Both are now unit-normalised before plotting, and the sign ambiguity is resolved before the cosine table is written.

Hedging Strategies I

Delta-Only

The control hedge removes underlying swap delta and leaves vega to run. It is useful as a lower bound because it captures the cost of pretending the cube is not the main risk. Delta P&L should be small in expectation, but vega and cross-vega dominate.

Single-Tenor Vega-Flat

Each new trade's vega is snapped to the nearest cube node and offset by trading that node. This looks clean in a blotter and can be deeply wrong in factor space if the offset node loads on the wrong combination of level, slope, and curvature.

Bucketed Vega-Flat

Bucketed hedging nets vega within expiry buckets: <=1y, 1y-3y, 3y-7y, and >7y. This is a practical compromise. It cannot fully neutralise factor exposure, but it stops the most obvious front/back leakage and uses fewer hedge instruments.

Portfolio Greek Matrix

Let V be node vega and L be the loading matrix. Raw vega reports V. Factor risk is closer to L'V, augmented by a skew exposure. The hedging question is therefore not only whether sum(V)=0, but whether the projected vector is small.

Hedging Strategies II

Factor-Neutral Hedge

The factor-neutral rule selects liquid cube points and solves a small constrained hedge to bring level, slope, curvature, and skew exposures near zero. It minimises residual factor exposure subject to trading-cost and instrument-size penalties.

$$\min_h ||A(q+Bh)||^2 + \lambda ||h||^2 + c^\top |h|$$

In trader language: use the liquid points that actually load on the factors you are trying to offset, and avoid over-trading an ill-conditioned local hedge.

Instrument Selection

The available hedge set is smaller than the risk grid. That is intentional. A market maker normally has a handful of executable liquid points and must project the book onto that basis. The factor-neutral hedge is therefore a practical proxy hedge, not a fantasy where every node trades frictionlessly.

Cost Accounting

Every hedge trade pays a configurable bid-ask cost in vol bp. The attribution table keeps that cost separate from cross-vega residuals, so the reader can see whether the tighter distribution was bought with too much turnover.

Market-Maker Simulation

Client Flow

Client trades arrive from a Poisson process. Cube point, strike offset, side, and notional are sampled from simple distributions. The client crosses the quote with probability near one. That simplification keeps the experiment focused on inventory and hedging rather than on a separate client-response model.

Book State

After each trade, inventory Greeks and factor exposures update. Hedging fires at end-of-day. The simulation records the book before and after hedging so P&L can be attributed to edge, theta, delta, first-order vega, cross-vega, and hedge cost.

Monte Carlo Scale

The default run uses 1,000 market-making paths over 252 trading days. Tests use small deterministic seeds and avoid Monte Carlo. Generated simulations are cached under outputs/simulations, while make clean removes them to prove reproducibility.

Simplifications

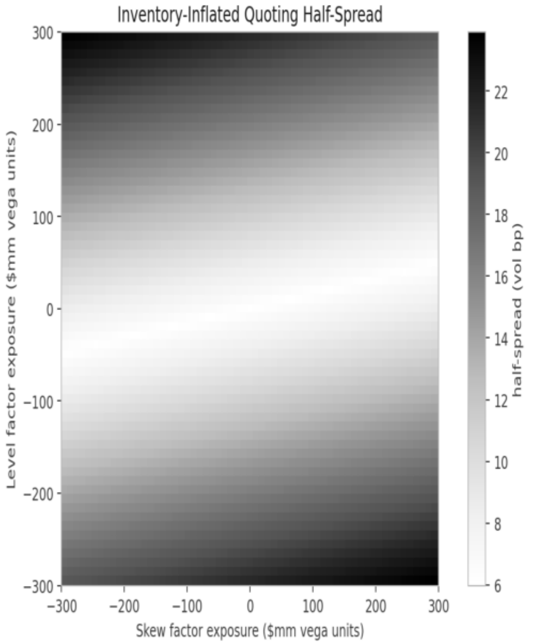
There is no adverse-selection model, no funding spread, no exchange margin, and no jump risk. Those choices make the experiment narrower, but they also make the hedge comparison easier to audit.

Quoting Overlay

Reservation Vol in Factor Space

The quote mid is adjusted by an Avellaneda-Stoikov-style inventory charge, but the inventory vector is factor exposure rather than a single scalar position.

$$r_i = mid_i - \gamma q^\top \Sigma l_i (T_{max} - t)$$



The heatmap is intentionally no longer flat. The displayed grid spans a wide enough inventory range for the risk-aversion term to matter, and the covariance matrix is used with off-diagonal terms intact. The gradient is therefore not purely axis-aligned. This remains a heuristic overlay, not a full dynamic-programming derivation.

Figure 6. Half-spread responds to level and skew inventory together.

Results I: Representative Path

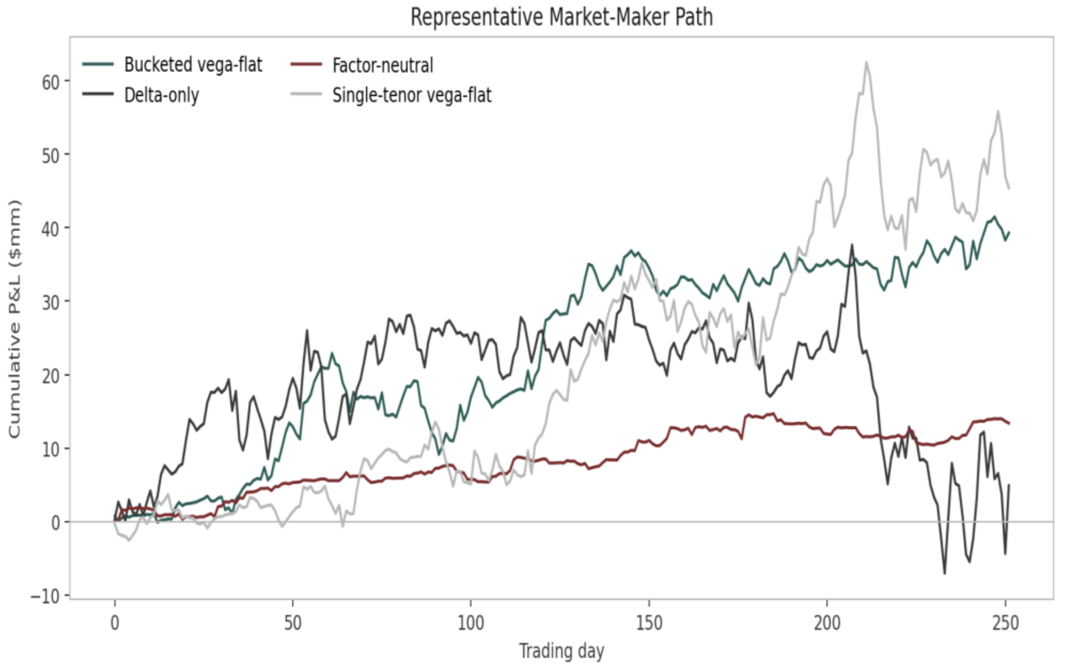


Figure 7. One path shows how the same client flow diverges under different hedges.

Path-Level Read

A single path should not be over-interpreted, but it is useful for sanity. The strategies see the same simulated flow and cube path. Divergence comes from the hedge state, hedge costs, and residual factor exposures rather than from different market draws.

Results II: Distributions and Tail Risk

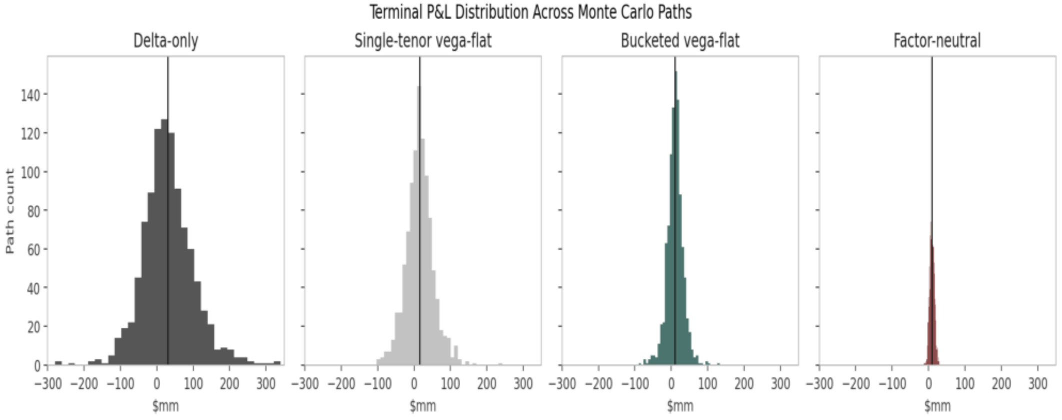


Figure 8. Small multiples keep the same x-axis while showing each strategy separately.

Tail-Risk Table ($mm)

Strategy	1% VaR	5% VaR	5% CVaR	Std
Delta-only	-121.4	-77.5	-115.1	70.6
Single-tenor vega-flat	-78.9	-46.1	-62.8	37.5
Bucketed vega-flat	-55.0	-25.0	-41.3	22.3
Factor-neutral	-4.4	-0.5	-2.8	6.4

The same-axis overlay in the executive summary is the headline picture. These small multiples make the same point strategy by strategy: delta-only carries the broadest distribution, bucketed hedging improves the shape, and factor-neutral hedging is the tightest in this experiment.

Results III: P&L Attribution

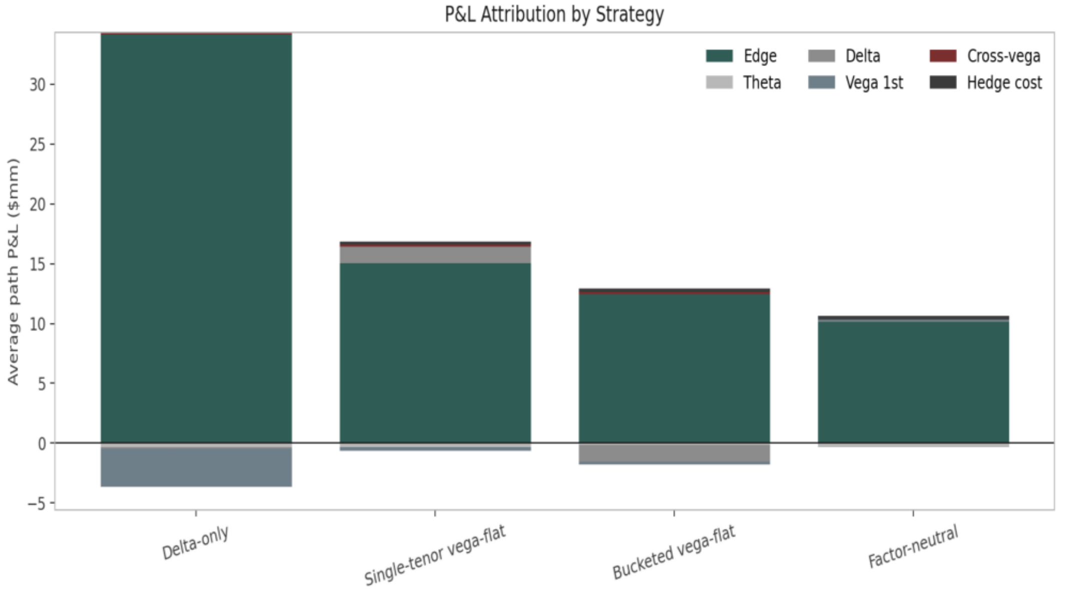


Figure 9. Attribution reconciles spread edge, Greeks, residual cross-vega, and cost.

Trader-Tone Read

The factor-neutral hedge does not create edge; it protects the edge already earned. Its cost is visible and should be challenged. The reason to like it is that the cross-vega residual is less dominant, so the P&L is less dependent on being lucky about which cube factor moved after the trade.

Results IV: What the Experiment Says

Headline

The result supports the core thesis in the controlled setting: a hedge constructed in the cube's factor basis produces a tighter realised P&L distribution than a hedge constructed only in local node vega. The improvement is clearest in the left tail and in the residual attribution.

Calibration Link

The calibration section tells the same story in model space. Per-slice fitting wins the local residual contest. The cube-constrained fit is the better surface object because it removes static-arbitrage breaks under a symmetric checker.

Caveat

This is not a live trading backtest. It is a market-making lab. The aim is to make the mechanics transparent enough that a reviewer can disagree with the parameters, rerun the project, and see how the result changes.

Desk Use

The immediate desk lesson is not to throw away node vega. Keep it for operations, but report the factor projection beside it. A position that is node-flat and factor-long should not be called flat.

Assumptions and Limitations

Simulated data only. No conclusion depends on a hidden vendor feed, but absolute P&L levels are not live-market forecasts.

Single-curve discounting. The lab ignores multi-curve basis and collateral details that would matter in production pricing.

Fixed beta. SABR beta is configurable but not calibrated; this keeps the exercise focused on cross-cube vega.

No jumps. Daily OU dynamics miss event risk, central-bank gap moves, and regime shifts.

Daily hedging. Intraday inventory management is compressed into an end-of-day hedge.

Perfectly observed mids. There is no mark uncertainty or broker-quality hierarchy.

Heuristic quoting. The factor-space reservation vol borrows the Avellaneda-Stoikov shape but is not a full HJB solution.

Extensions

Real-data overlay

Use exchange settles or broker indicative marks to initialise the cube and compare factor stability.

Rough-vol dynamics

Replace OU factors with rougher latent drivers and test whether hedging cadence becomes more important.

Multi-curve pricing

Add OIS discounting and forward curves so the swaption lab better resembles production infrastructure.

Joint cap-swaption calibration

Bring caplets into the smile calibration and ask whether short-rate consistency changes the hedge basis.

Bermudan extension

Use least-squares Monte Carlo to push the factor hedge into callable-exotics inventory.

Strategic quoting

Let IDB visibility and client response feed back into the reservation-vol rule.

Learning overlay

Use reinforcement learning as a policy layer on top of the transparent factor state, not as a black-box replacement.

References I

Andersen and Piterbarg (2010)

Interest Rate Modeling. Atlantic Financial Press. The project borrows its interest-rate modelling discipline from this reference: explicit curves, model assumptions, and no mystery calibration knobs.

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References II

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High-frequency trading in a limit order book. Quantitative Finance 8(3), 217-224. The quote overlay borrows the reservation-price intuition, with the caveat that this project adapts it heuristically to vol-space factors.

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Interest Rate Models: Theory and Practice. Springer. Background for rate derivatives, annuities, and swaption pricing conventions.

Glasserman (2004)

Monte Carlo Methods in Financial Engineering. Springer. Background for deterministic simulation design, reproducibility, and path-level diagnostics.

Build Note

The sandbox used for this build did not provide XeLaTeX or dvisvgm. The repository still writes full LaTeX sources and a TikZ architecture diagram, while the delivered PDF is rendered through a deterministic Matplotlib fallback with the same palette.